

Thermal Measurements and Geothermal Gradient from Deep Boreholes in the Revell Batholith, Northwest Ontario

Kasani, H.A., DesRoches, A., Sykes, E., Parmenter, A. and Khorshidi, M.

Nuclear Waste Management Organization (NWMO), Toronto, Ontario, Canada

Copyright 2024 ARMA, American Rock Mechanics Association

This paper was prepared for presentation at the 58th US Rock Mechanics/Geomechanics Symposium held in Golden, Colorado, USA, 23-26 June 2024. This paper was selected for presentation at the symposium by an ARMA Technical Program Committee based on a technical and critical review of the paper by a minimum of two technical reviewers. The material, as presented, does not necessarily reflect any position of ARMA, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper for commercial purposes without the written consent of ARMA is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 200 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgement of where and by whom the paper was presented.

ABSTRACT: The Revell Site, in northwestern Ontario, within the Canadian Shield, is presently being studied as one of the sites for hosting a deep geological repository (DGR) in Canada. Preliminary site investigation included the drilling and coring of six 1-km long boreholes. The thermal characterization involved obtaining temperature logs from multiple sources in boreholes and laboratory thermal properties testing of different rock types. After drilling, coring, and flushing of the borehole, downhole temperature data were collected in this order: (1) continuous logs from geophysical dynamic fluid temperature probe, (2) discrete temperature data measured during hydraulic packer testing, and (3) discrete temperature data collected post installation of a Westbay monitoring system. The hydraulic testing and Westbay data provided the most reliable estimates of in-situ temperature profile with an average geothermal gradient of 9.9°C/km. This gradient is within the lower range of the data from other study sites in the Canadian Shield. The low gradient is consistent with low concentrations of natural uranium, thorium, and potassium in the tested core samples, which results in a low radiogenic heat generation rate. This paper presents a summary of the thermal data collection in the first three boreholes and development of a thermal model for the site, which provides boundary conditions for thermal simulations of the DGR. This initial thermal property dataset and site model also provides a prediction against which data from the latter three boreholes can be expected.

1. INTRODUCTION

Deep geological disposal is an internationally accepted method for the safe and long-term management of used nuclear fuel. Safe long-term performance of a Deep Geological Repository (DGR) spans over the construction, operation, and post-closure phases which includes geological processes such as earthquakes and glacial advance and retreat.

The Nuclear Waste Management Organization (NWMO) is implementing Adaptive Phased Management (APM), the approach approved by the Government of Canada in 2007 for the long-term management of used nuclear fuel. The NWMO site selection process is described in NWMO (2010). As part of APM, the NWMO is conducting preliminary site assessment activities in potential siting areas in: i) crystalline rocks in the northern portion of the Revell batholith within the Canadian Shield, near the Township of Ignace in the traditional territory of Wabigoon Lake Ojibway Nation (WLON) in northwest Ontario (referred to as the Revell Site), and ii) in sedimentary rocks in the Municipality of South Bruce in the traditional territory of Saugeen Ojibway Nation (SON) in southern Ontario (referred to as the South Bruce Site).

Preliminary site investigation activities at the Revell Site included the drilling, coring, and testing of six 1000-m long boreholes (IG_BH01 to IG_BH06) with extensive downhole measurements and laboratory testing. Given that high-level used fuel emits heat over a long period of time, initial thermal characterization activities were also conducted. The overall objective and scope of the thermal characterization program was to assess and summarize key thermal parameters of the bedrock at the Revell Site, where intersected by deep boreholes. The thermal characterization included obtaining temperature logs from multiple sources in one vertical borehole and five inclined boreholes, laboratory determination of thermal properties (thermal conductivity, thermal diffusivity, volumetric heat capacity, linear thermal expansion coefficient), and mineralogical compositions of samples of different rock types present in the boreholes. Thermal characterization provides information on thermal boundary conditions, geothermal gradient, and the distribution of thermal properties in the crystalline bedrock, which will be used for simulation of temperature evolution of the DGR, safety assessments, and repository layout design.

In the past, the NWMO has conducted modeling of thermal performance of a DGR in a hypothetical site within the Canadian Shield (Guo, 2023; Radakovic-Guzina et al., 2023). After obtaining site-specific data

including thermal data, currently additional studies are ongoing to investigate thermo-hydro-mechanical performance of a potential DGR in the Revell Site.

This study presents a summary of the thermal data collection in the first three boreholes while interpretation and integration of the data from the other three boreholes are ongoing.

2. GEOLOGY OF THE REVELL SITE

The Revell batholith is roughly rectangular in shape, trends northwest, is approximately 40 km in length, and covers an area of approximately 455 km² (Fig. 1). Four main suites of plutonic rock are recognized in the Revell batholith, including, from oldest to youngest: biotite tonalite to granodiorite suite, a hornblende tonalite to granodiorite suite, a biotite granite to granodiorite suite, and a K-feldspar megacrystic granite. Plutonic rocks of the biotite tonalite to granodiorite suite occur along the southwestern and northeastern margins of the Revell batholith. The principal type of rock within this suite is a white to grey, medium-grained, variably massive to foliated or weakly gneissic, biotite tonalite to granodiorite. Detailed summary of Revell batholith can be found in Parmenter et al. (2020).

Szewczyk and West (1976) interpreted the Revell batholith to be a sheet-like intrusion that is approximately 1.6 km thick. More recent 2.5D gravity modeling suggested that the batholith is on the order of 2 km to 3 km thick through the center of the northern part of the batholith (SGL, 2020).

Based on results from six 1000-m long boreholes (IG_BH01 to IG_BH06) the Revell batholith is extremely homogeneous, with biotite granodiorite-tonalite to biotite tonalite representing more than 97% of the bedrock encountered, with the remaining comprised of felsic dykes and amphibolite. Though minor by volume, the amphibolite occurrences in all boreholes represent marker units that are useful to subdivide the otherwise homogeneous bedrock into distinct, gently dipping rock units.

3. PRELIMINARY SITE INVESTIGATION

3.1. Deep Borehole Drilling

Six boreholes were drilled using a diamond drill in the Revell batholith, each 1000 m along their length. Borehole IG_BH01 was planned as a vertical borehole,

while the other five boreholes were drilled at a planned inclination of 70° with different azimuths. The boreholes were installed with steel casing down to approximately 60 to 70 m in all boreholes.

3.2. Downhole Temperature Data Collection Methods

After the drilling and coring of the borehole was completed and the borehole was flushed with fluorescein traced drillwater, downhole temperature data were collected in the following order:

- (i) Geophysical Dynamic Fluid Temperature Probe Continuous temperature data was collected from the geophysical probe in all three boreholes.

- (ii) Field Hydraulic Packer Testing Discrete temperature data were collected during the hydraulic packer testing in all three boreholes.

- (iii) Westbay Monitoring System Discrete temperature data were collected using Westbay MP38 multi-level groundwater monitoring system installed in boreholes IG_BH01 and IG_BH03.

3.3. In-Situ Temperature Profile in Boreholes

Field temperature data measured by the geophysical probe, during hydraulic testing, and using the Westbay system in IG_BH01 are shown in Fig. 2. The Westbay temperature data were collected on multiple dates. The temperature data from the geophysical probe and Westbay system are in good agreement. As depicted in Fig. 2, temperature at any specific depth remained nearly constant for the duration of the data collection.

Fig. 3 presents the field temperature data versus true vertical depth from the geophysical probe and during hydraulic testing in IG_BH02. The data from the geophysical probe were higher than those measured during the hydraulic testing.

Temperature data from the geophysical probe, during hydraulic testing, and from the Westbay system in IG_BH03 are plotted against true vertical depth in Fig. 4. The Westbay temperature data were collected on multiple dates. As shown in Fig. 4, temperature remained nearly constant for the duration of the data collection by hydraulic testing and the Westbay system. The geophysical probe temperature data were higher than temperature data from the hydraulic testing and Westbay system.

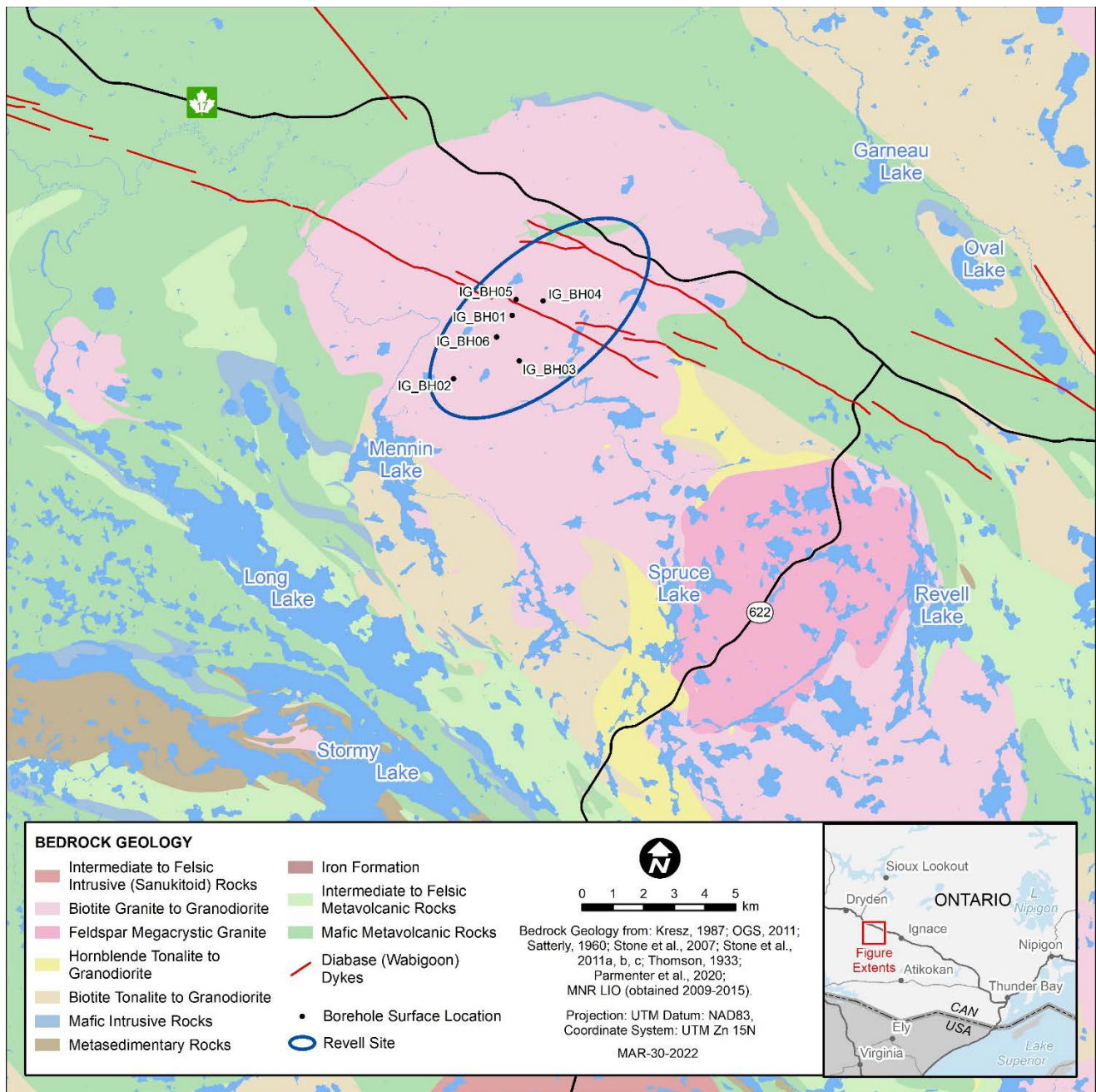


Fig. 1. The Revell Site (shown in the blue oval) in the surface bedrock map of the Revell area and locations of IG_BH01 to IG_BH06 (modified from Parmenter et al., 2020).

4. GEOTHERMAL GRADIENT FOR THE REVELL SITE

Fig. 5 presents the combined suite of all temperature data versus true vertical depth, in all three boreholes. The temperature data recorded by the geophysical probe in IG_BH01 closely followed other IG_BH01 datasets. Conversely, the temperature readings from the

geophysical probe in IG_BH02 and IG_BH03 were consistently higher than those from the hydraulic testing and Westbay systems. Both sets of data from the hydraulic testing in IG_BH02 and IG_BH03 and the Westbay system in IG_BH03 recorded slightly higher temperatures than the data recorded for IG_BH01 up to a depth of about 500 m; however, below this depth, all measurements are approximately aligned.

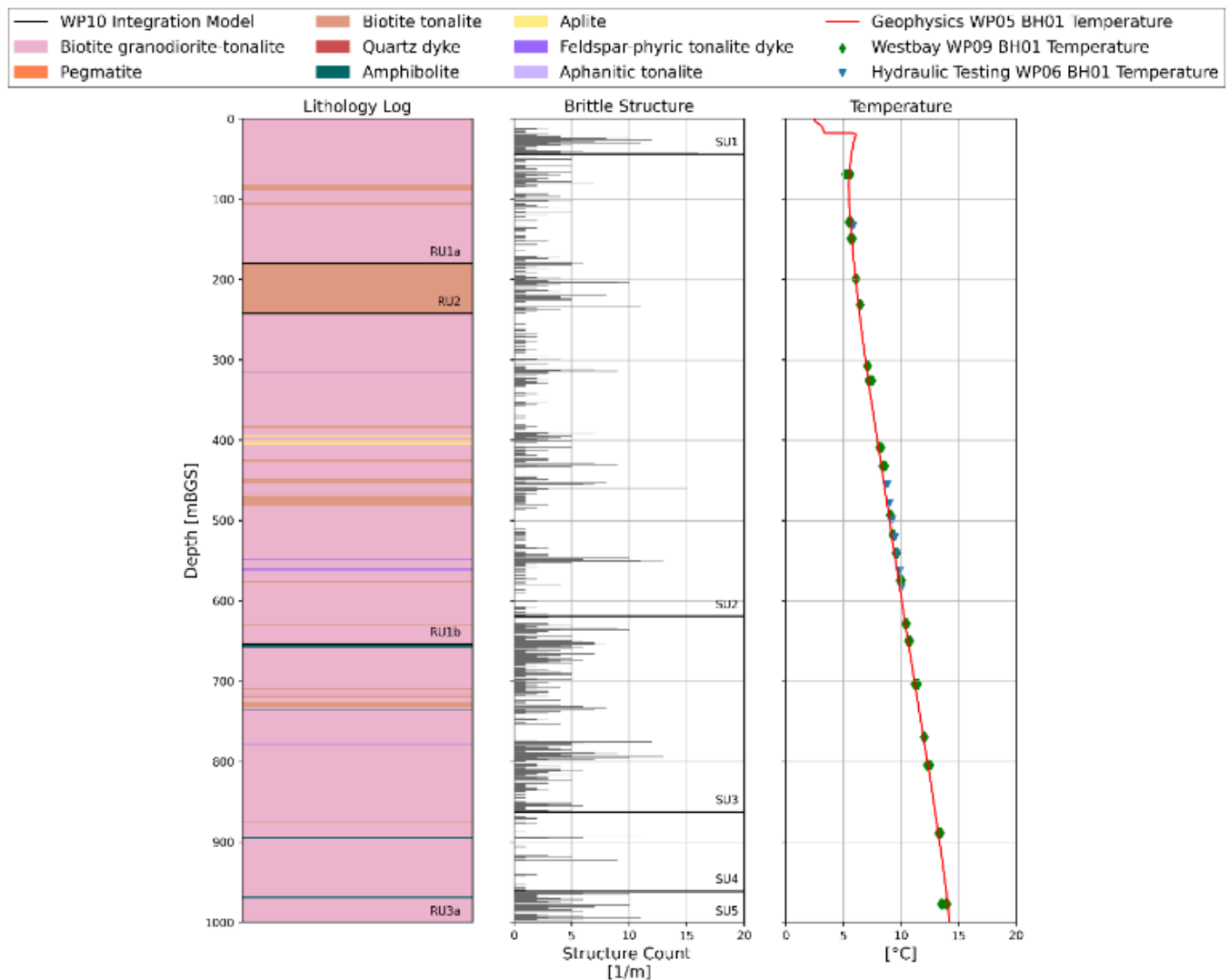


Fig. 2. Temperature-depth data measured by the geophysical probe, during hydraulic testing, and with the Westbay system in IG_BH01. The temperature logs are accompanied by the lithology and brittle structure logs.

All three geophysical probe temperature datasets in IG_BH01, IG_BH02, and IG_BH03, showed inflections in their profiles, indicating minimum temperatures around 100 m vertical depth. Drilling fluid temperature affects the temperature of the rocks surrounding the borehole; also flow or lack of flow into the boreholes may impact their ability to come to equilibrium. Therefore, it is possible that the temperatures recorded by the geophysical probes were impacted by the temperature of the drilling fluid, which itself depends on the seasonal weather temperature. IG_BH01 was drilled in the winter months, with a resulting lower temperature drilling fluid, while IG_BH02 and IG_BH03 were drilled in the summer

and fall, respectively, with resulting higher temperature drilling fluids. The other observation is that the temperature recorded by the geophysical probe from the near surface depths in all boreholes clearly reflected disturbance by air temperature. Overall, based on the likely seasonal influence of drilling fluid temperature on the geophysical probe temperature measurements, it is concluded that the hydraulic testing and Westbay data should be regarded as the most reliable sources of in-situ temperature data for the boreholes since sufficient time since drilling and flushing has occurred for the fluid temperatures to equilibrate with the geothermal gradient.

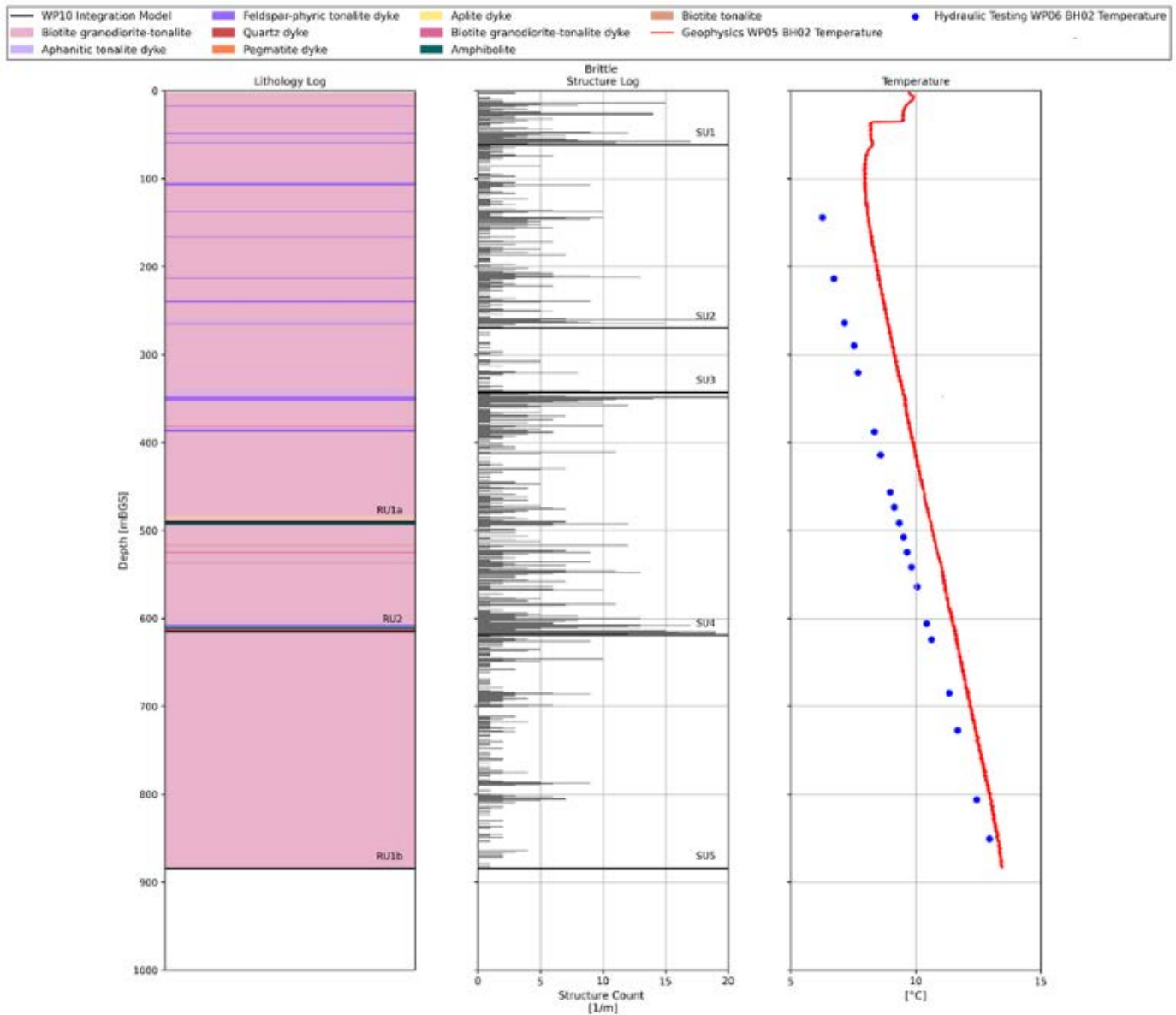


Fig. 3. Temperature versus true depth data measured by the geophysical probe and during hydraulic testing in IG_BH02. The temperature logs are accompanied by the lithology and brittle structure logs.

Geothermal gradients from the hydraulic testing in IG_BH01, IG_BH02, and IG_BH03 were determined as 9.44, 9.60, and 9.63°C/km, respectively. The Westbay system data resulted in geothermal gradient of 10.05 and 9.40°C/km in IG_BH01 and IG_BH03, respectively. However, in order to determine an average geothermal

gradient for the Revell Site, all temperature data collected during the hydraulic testing and on multiple dates by the Westbay system are plotted in Fig. 6. Between the shallowest and deepest measurements depth (i.e., 69.1 m and 977.3 m, respectively), a linear regression line was fitted with a regression coefficient $R^2 = 0.987$.

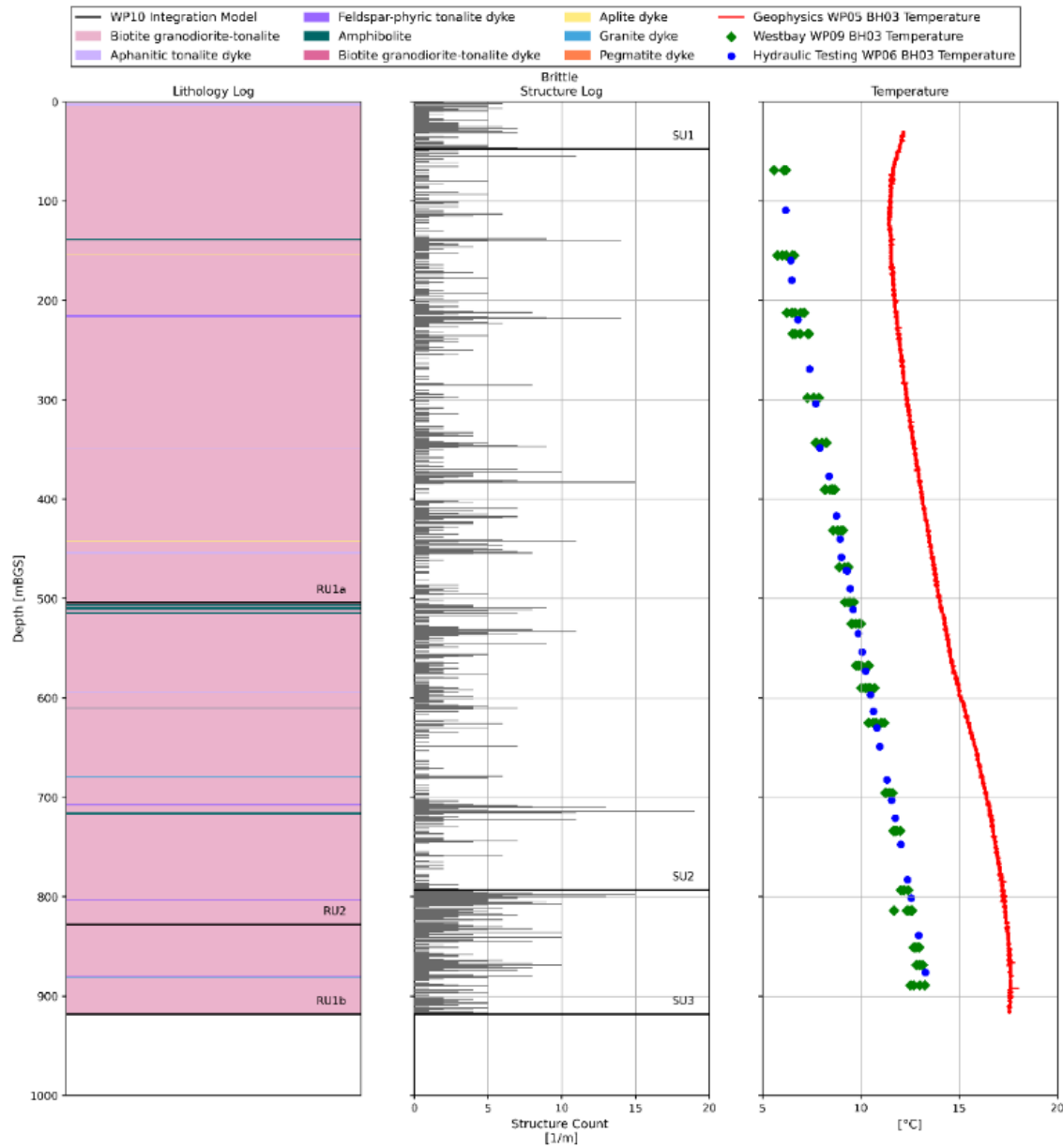


Fig. 4. Temperature versus true depth data measured by the geophysical probe, during hydraulic testing, and with the Westbay system in IG_BH03. The temperature logs are accompanied by the lithology and brittle structure logs.

The data in Fig. 6 shows some scatter in temperature data. This may suggest that there is a minimum in the temperature-depth profile at around the shallowest data point depth. The inflections in the profiles from all geophysical temperature in the three boreholes are located at around 100 m depth (Fig. 5). Drury and Lewis (1983) also reported a minimum temperature at a depth of 60-70 m in boreholes drilled in the Lac du Bonnet batholith of the Superior Province in the Canadian Shield. Therefore, it is assumed that the minimum temperature in IG_BH01, IG_BH02, and IG_BH03 is at a depth of 69.1 m and, hence, due to the lack of reliable data from shallow depth (and especially given the installation of steel casing down to a depth of 60 to 70 m), it is suggested to apply a constant temperature of 5.1°C for the depth range from ground surface to 69.1 m, which is equal to the value predicted by the best linear fit for a depth of 69.1 m.

Therefore, the following piecewise function for temperature versus depth is proposed for the Revell site. Based on this function, for the depth range of 69.1 m to 977.3 m, the average geothermal gradient is 9.9°C/km.

$$T = 5.1; \text{ if } D < 69.1 \text{ m} \quad (1)$$

$$T = 0.0099D + 4.454; \text{ if } 69.1 < D < 977.3 \text{ m} \quad (2)$$

where T is temperature in °C and D is vertical depth in meters below ground surface.

The relatively linear temperature-depth profile at the site as depicted in Fig. 6 suggests that there are no thermal anomalies present beneath the site at the scale of rock units or structural units. It is also worth noting, however, that there is a minor degree of scatter in the temperature data at a more local scale. At this stage, it is unclear

whether this variability is due to resolution of the instruments or real geological character of the site.

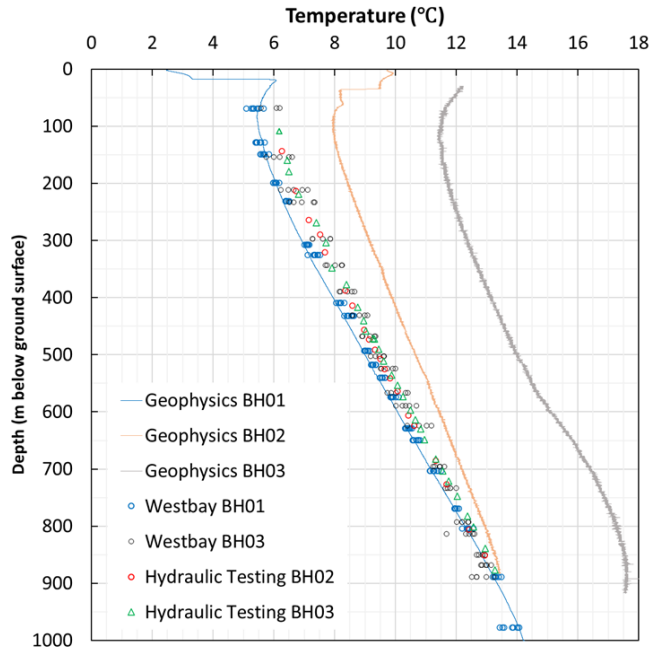


Fig. 5. Borehole temperature data versus true vertical depth from the geophysical probe, hydraulic testing, and Westbay system in IG_BH01, IG_BH02, and IG_BH03.

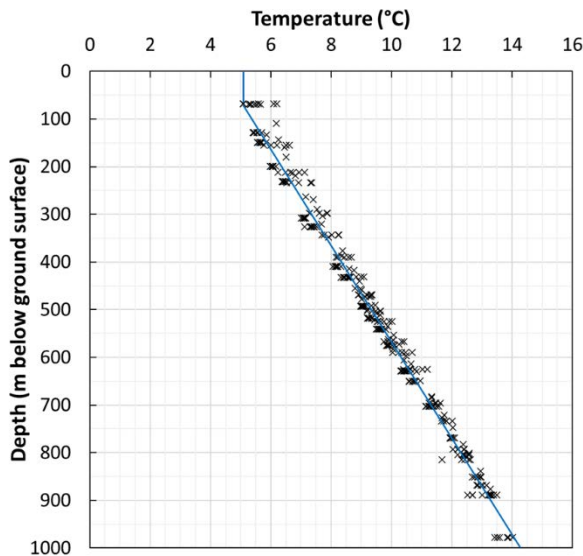


Fig. 6. Integrated temperature-depth data from the hydraulic testing and Westbay data, and the piecewise function for the average temperature gradient for the Revell Site (Equation 1 and 2 in the text).

5. LABORATORY THERMAL PROPERTIES TESTING

Laboratory thermal properties testing consisted of determination of thermal conductivity, thermal diffusivity, and volumetric heat capacity measurements using a hot disc equipment according to ISO (2015) and linear thermal expansion coefficient following ASTM

(2013), on core samples from IG_BH01 and IG_BH03. Thermal conductivity of all samples measured at different temperatures, between about 25 and 99°C, are shown in Fig. 7. Similarly, thermal diffusivity and volumetric heat capacity values were determined for all samples at all temperatures (not shown here). Thermal conductivity and diffusivity generally decreased with temperature whereas volumetric heat capacity generally increased with temperature.

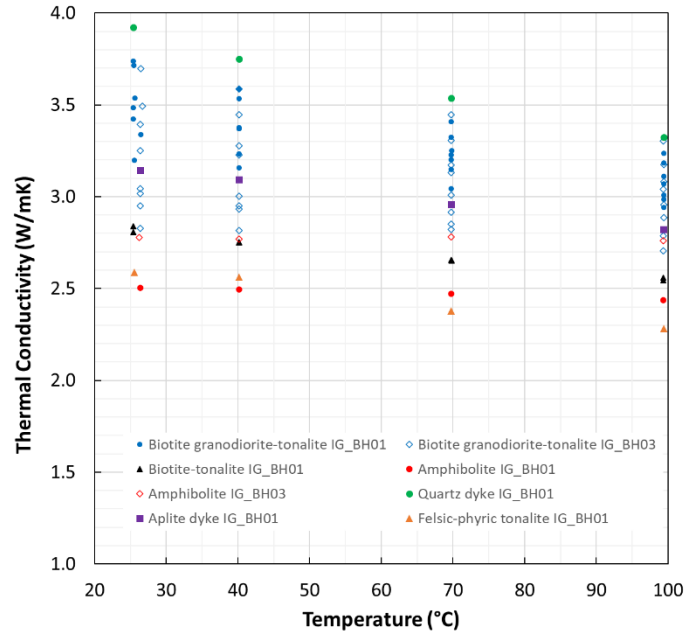


Fig. 7. Thermal conductivity of samples from all rock type versus temperature. IG_BH01 samples were collected from the core boxes post-drilling and re-saturated prior to the testing by submersion into water under vacuum. IG_BH03 samples were collected during the drilling, preserved, and tested under as-received water content conditions.

Table 1 presents mean thermal properties of all rock types at room temperature. Mineralogical testing showed that the biotite granodiorite-tonalite rock samples, on average, consisted of 40% quartz compared to the amphibolite samples which contained less than 4% quartz. Different rock-forming minerals have different thermal conductivity; thermal conductivity of quartz is the highest and equal to 7.69 W/mK (Horai and Simmons, 1969). This explains why the sample from the quartz dyke in Fig. 7 and Table 1 showed the highest thermal conductivity. Similarly, given the higher quartz content, the biotite granodiorite-tonalite rock samples showed higher thermal conductivity than the amphibolite samples.

Table 1. Mean thermal properties of samples of all rock types at room temperature, from IG_BH01 and IG_BH03

Rock type ¹	Mean thermal conductivity (W/mK)	Mean thermal diffusivity (mm ² /s)	Mean volumetric Heat capacity (MJ/m ³ K)
Biotite granodiorite-tonalite (16)	3.34	1.68	2.00
Amphibolite (2)	2.64	1.16	2.29
Biotite-tonalite (2)	2.83	1.36	2.08
Quartz dyke (1)	3.92	1.96	2.00
Aplite dyke (1)	3.14	1.70	1.85
Feldspar phyric tonalite dyke (1)	2.59	1.24	2.09

¹ Number in parentheses indicate total number of samples from the two boreholes.

Coefficient of linear thermal expansion of samples from IG_BH01 and IG_BH03 were determined between about 22 and 90°C. Two heating-cooling cycles were conducted; no noticeable differences between thermal expansion coefficients from the first and second cycles were found. All sample thermal strain linearly increased with temperature in this temperature range (see an example plot in Fig. 8). Table 2 presents mean linear thermal expansion coefficients of all rock types which are very close to each other.

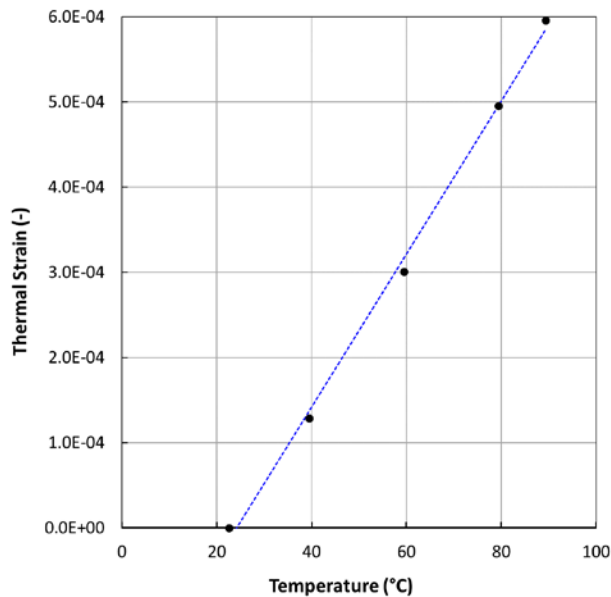


Fig. 8. Thermal strain of a re-saturated biotite granodiorite-tonalite sample from IG_BH01 as a function of temperature, during first heating path. The linear best fit line results in a linear thermal expansion coefficient of 8.97E-06 (1/°C).

Table 2. Mean linear thermal expansion coefficient samples of all rock types between 25 and 90°C, from first heating path, from IG_BH01 and IG_BH03

Rock type ¹	Mean linear thermal expansion coefficient ($\times 10^{-6}$ 1/°C)
Biotite granodiorite-tonalite (16)	7.48
Amphibolite (2)	7.75
Biotite-tonalite (2)	7.05
Aplite dyke (1)	5.88
Feldspar phyric tonalite dyke (1)	6.45

¹ Number in parentheses indicate total number of samples from the two boreholes.

6. COMPARISON WITH OTHER STUDY SITES IN THE CANADIAN SHIELD AND INTERNATIONAL DATA

6.1. Geothermal Gradient

The geothermal gradient observed at the Revell site is at the lower end of the range of data from other sites including AECL sites in Canada, SKB's Forsmark Site in Sweden, and Posiva's Olkiluoto Site in Finland data (Table 3). SKB's and AECL's sites share similarities with the Revell Site as they are all Archean (AECL sites) or Paleoproterozoic (Forsmark Site) cratonic environments with granitoid host rocks. The reported gradients for AECL sites were between 9.9 and 14.4°C/km. The temperature gradient in the boreholes drilled by SKB at the Forsmark Site varies between 10.5 and 12.0°C/km whereas Posiva (2011) reported a gradient of 14.1°C/km for the depth from 350 to 500 m.

Table 3. Comparison of Geothermal Gradients at the Revell Site with AECL (Canada), SKB (Sweden), and Posiva (Finland)

Site		Geothermal Gradient (°C/km)
Revell Site (present study)		9.9
AECL (Snowden et al. 2021)	Atikokan	11.24
	Chalk River	10.55
	Lac du Bonnet Batholith/URL	9.9-14.4
	Whiteshell	10.09
Forsmark Site (SKB, 2007)		10.5~12.0
Olkiluoto Site (Posiva, 2011)		14.1 ¹

¹ For the depth range of 350-500 m

6.2. Laboratory Thermal Properties

Thermal conductivity, thermal diffusivity, and heat capacity of the rocks from the Revell Site are within the values reported for the Lac du Bonnet Batholith (Drury and Lewis, 1983), the Forsmark Site (SKB, 2007), and the Olkiluoto Site (Posiva, 2011). The trends of these properties with temperature follow those from the literature (e.g., Heuze, 1983; SKB, 2007; Vosteen and Schellschmidt, 2003; Zhao et al., 2016).

The thermal expansion data from the Revell site are also comparable to the Forsmark Site (SKB, 2007) and international data (e.g., Heuze, 1983).

6.3. Radiogenic Heat Generation Rate

The low geothermal gradient at the Revell Site is consistent with very low concentrations of natural uranium, thorium, and potassium in the tested core samples which results in a low radiogenic heat generation rate. The radiogenic heat generation rate was determined using Equation (3) based on the concentrations of uranium (C_U) and thorium (C_{Th}) in ppm unit and the concentration of potassium (C_K) in weight percent as proposed by Rybach (1988).

$$H_r = 10^{-2} \rho (9.52C_U + 2.56C_{Th} + 3.48C_K) \quad (3)$$

where H_r is the radiogenic heat generation rate in $\mu\text{W}/\text{m}^3$ and ρ is the average density of the rock in g/cm^3 .

Based on whole rock geochemistry analysis on 93 samples from boreholes IG_BH01, IG_BH02, and IG_BH03 from the Revell Site, the average values for the concentration of uranium, thorium, and potassium are 2.081 ppm, 5.247 ppm, and 2.079%, respectively. With an average density of $2.66 \text{ g}/\text{cm}^3$ for the biotite granodiorite-tonalite and biotite tonalite (the main rock consisting about 97% of the drilled boreholes), the radiogenic heat generation rate for the Revell Site is estimated to be $1.08 \mu\text{W}/\text{m}^3$. This value is lower than the rate for the Lac du Bonnet Batholith ($5.23 \mu\text{W}/\text{m}^3$) as reported by Drury and Lewis (1983).

7. CONCLUSIONS

The thermal data summary provides information on in-situ thermal boundary conditions, in-situ geothermal gradient, and thermal characterization of the rock type present at the Revell Site. Based on the current study, the following conclusions can be drawn:

- The field hydraulic testing and the Westbay system provide reliable estimates of in-situ temperature profile. The average geothermal gradient from these two data source results in a geothermal gradient of $9.9^\circ\text{C}/\text{km}$ for the Revell Site. A piecewise function is proposed; a constant temperature values of 5.1°C is suggested for 0-69.1 m depth, and a linear function with depth for 69.1-977.3 m can be considered using an average gradient of $9.9^\circ\text{C}/\text{km}$.
- Concentration of uranium, thorium, and potassium result in an approximate radiogenic heat generation rate of $1.08 \mu\text{W}/\text{m}^3$. The geothermal gradient at the Revell site is at the lower end of the range of data from AECL, Posiva, and SKB sites. This is consistent with the

low radiogenic heat generation rate at the Revell Site compared to the Lac du Bonnet batholith.

- Laboratory thermal properties of all rock types from the Revell Site are comparable with international data.

REFERENCES

- ASTM. (2013). Standard Test Methods for Measurement of Thermal Expansion of Rock Using Dilatometer. ASTM D4535-13e2.
- Drury, M. J., and Lewis, T. J. (1983). Water movement within Lac du Bonnet Batholith as revealed by detailed thermal studies of three closely-spaced boreholes. *Tectonophysics*, 95(3-4), 337-351.
- Guo, R. (2023). Calculation of thermal-hydraulic-mechanical response of a deep geological repository for radioactive used fuel in granite. *Int. J. Rock Mech. Min. Sci.*, 170, 105435.
- Heuze, F. E. (1983). High-temperature mechanical, physical and thermal properties of granitic rocks—a review. *Int. J. Rock Mech. Min. Sci. Geomech. Abstr.* 20(1), 3-10.
- Horai, K. I., and Simmons, G. (1969). Thermal conductivity of rock-forming minerals. *Earth and Planetary Science Letters*, 6(5), 359-368.
- ISO. (2015). *Plastics – Determination of thermal conductivity and thermal diffusivity*, Part 2: Transient plane heat source (hot disc) method. ISO 22007-2:2015 (E).
- NWMO. (2010). *Moving Forward Together: Process for Selecting a Site for Canada's Deep Geological Repository for Used Nuclear Fuel*. Toronto, Ontario, Canada.
- Parmenter, A., Waffle, L., and DesRoches, A. (2020). *Bedrock Geology of the Revell Batholith and Surrounding Greenstone Belts*. Nuclear Waste Management Organization Report NWMO-TR-2020-08. Toronto, Canada.
- Posiva. (2011). *Olkiluoto Site Description*. Posiva 2011-02.
- Radakovic-Guzina, Z., Damjanac, B., Lam, T., and Kasani, H. A. (2023). Numerical simulation of long-term performance of deep geological repository placement rooms in crystalline and sedimentary rocks. *Comput. Geotech.*, 157, 105348.
- Rybach L. (1988). Determination of heat production rate. In: Haenel R., Rybach L., Stegena L., editors. *Handbook of Terrestrial Heat Flow Density Determination*, pages 125-142. Kluwer Academic Publishers Group, Dordrecht.
- SGL. (2020). *3D Geophysical Forward and Inversion Modelling of the Revell Batholith and Surrounding Greenstone Belt*. Nuclear Waste Management Organization, Toronto, Ontario, Canada. Report APM-REP-01332-0270.
- SKB. (2007). *Thermal properties, Site Descriptive Modelling Forsmark – Stage 2.2*. R07-47.
- Snowdon, A. P., Normani, S. D., and Sykes, J. F. (2021). Analysis of crystalline rock permeability versus depth in a Canadian Precambrian rock setting. *J. Geophys. Res. Solid Earth*, 126(5), e2020JB020998.

Szewczyk, Z. J., and West, G. F. (1976). Gravity study of an Archean granitic area northwest of Ignace, Ontario. *Can. J. Earth Sci.*, 13(8), 1119-1130.

Vosteen, H. D., and Schellschmidt, R. (2003). Influence of temperature on thermal conductivity, thermal capacity and thermal diffusivity for different types of rock. *Phys. Chem. Earth*, Parts a/b/c, 28(9-11), 499-509.

Zhao, X. G., Wang, J., Chen, F., Li, P. F., Ma, L. K., Xie, J. L., and Liu, Y. M. (2016). Experimental investigations on the thermal conductivity characteristics of Beishan granitic rocks for China's HLW disposal. *Tectonophysics*, 683, 124-137.